

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10: VOLUME AND CAPACITY (SUB-STRAND 2.7)

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation of the Kericho containers puzzle. Mr. Kiprop asked Mama Chebet why his bucket is not as big as her drum when they look the same size. You will now answer him using mathematics. Use $\pi = 22/7$ in all calculations. Remember: $1 \text{ cm}^3 = 1 \text{ mL}$, $1000 \text{ cm}^3 = 1 \text{ L}$ and $1 \text{ m}^3 = 1000 \text{ L}$. Write your answers in the space under each prompt. Show every step of your working, include units, and give litres to 2 decimal places for the Kericho containers. Refer to the containers as drum (A), cover (B) and bucket (C). Use the evidence you collected in Lessons 1 to 8 and link every answer back to the driving question.

Part 1: Same Look, Different Shape

Mama Chebet's drum (A), conical sisal cover (B) and metal bucket (C) are all about 60 cm tall with a 42 cm opening. (a) Write down the opening radius. (b) Explain why having the same height and the same opening does not mean the containers hold the same amount. (c) Think back to your Lesson 1 prediction. How did your predicted order compare with what you later found, and what idea were you missing at that time? (d) State the difference between volume and capacity, with the units used for each.

Write your answer here:

Part 2: The Drum and the Cover

(a) Describe the pouring investigation you did with a hollow cone and a hollow cylinder of the same base radius and height. What did you find? **(b)** Write the formula for the volume of a cone. **(c)** Calculate the volume of the cylindrical drum (A) and of the conical cover (B) in cm^3 and then in litres. **(d)** Show how your answers prove the relationship between the drum and the cover.

Write your answer here:

Part 3: Explaining the Bucket

Bucket C has top radius $R = 21$ cm, bottom radius $r = 14$ cm and height $h = 60$ cm. (a) Explain how a frustum of a cone can be thought of in terms of two cones. (b) Use similar triangles to find the height of the small cone that was removed and of the full cone. (c) Calculate the volume of the full cone and of the small cone. (d) Find the capacity of the bucket in litres and check it with the frustum formula. (e) Answer Mr. Kiprop: why is his bucket not as big as the drum, yet bigger than the cover? Use your numbers.

Write your answer here:

Part 4: Pyramids and Their Frustums

(a) A hollow square-based pyramid with base $30\text{ cm} \times 30\text{ cm}$ and height 40 cm was poured into a cuboid of the same base and height. State what you found and write the pyramid volume rule. Calculate the pyramid's volume in cm^3 and litres. (b) A maize bin for Rift Valley farmers is a square-based frustum with top side 60 cm , bottom side 30 cm and height 40 cm . Calculate its capacity in litres using the frustum formula for a pyramid. (c) Check your answer using the large-minus-small pyramid method. (d) The bulk density of maize is 0.75 kg per litre . Find the mass of maize the bin holds.

Write your answer here:

Part 5: Designing and Using Mathematics

(a) A cup is a frustum with top radius 6 cm, bottom radius 3 cm and height 7 cm. Calculate its volume, then find how many such cups fill the Kericho drum (A). (b) When you built and tested your own model containers, your measured capacity was not exactly the same as the calculated one. Explain how you calculated percentage error and give two reasons for the difference. (c) Write a short final explanation (4 to 6 sentences) that answers the driving question: why do the three containers hold such different amounts, and how can we use mathematics to design a frustum container that holds exactly the capacity we need? Include the key numbers. (d) Using your answer to the KICD key inquiry question, explain why knowledge of volume and capacity is useful.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Mathematical accuracy and units	All volumes, capacities, ratios and conversions are correct, use $\pi = 22/7$, and include correct units with litres to 2 decimal places for the Kericho containers.	Most calculations are correct, with one or two small slips in arithmetic or units that do not change the conclusion.	Several calculations or conversions are incorrect, or units are missing, so the conclusions are not supported.
Use of formulas and methods	Chooses the correct formula for each solid and uses the large-minus-small method and similar triangles accurately, showing the cone-cylinder and pyramid-cuboid one-third relationships.	Uses the correct formulas in most cases but the working for similar triangles or large-minus-small is partly incomplete.	Uses wrong formulas or cannot show how a frustum relates to a cone or pyramid.
Explaining the phenomenon	Clearly explains that the containers differ in capacity because their shape differs,	Explains that shape matters and quotes the capacities, but the reasoning about	States that the containers differ without a mathematical explanation or with

	uses 27.72 L, 58.52 L and 83.16 L, and answers Mr. Kiprop's question with the tapering reasoning.	tapering is general or incomplete.	incorrect numbers.
Evidence, error and design reasoning	Uses evidence from the pouring and model investigations, explains percentage error and its causes, and describes a clear process for designing a frustum of a required capacity.	Mentions evidence and at least one cause of error, and gives a partial design process.	Gives little evidence from the investigations and no clear explanation of error or design.
Communication and usefulness of volume and capacity	Writes clear, well-organised steps in sensible language and gives well-chosen Kenyan examples that show why volume and capacity are useful.	Writing is mostly clear and gives one or two examples of usefulness.	Writing is hard to follow and gives no examples of why volume and capacity are useful.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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